

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Histograms

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

HISTOGRAMS · CH6 · L18

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The Map

This lesson collapses Histograms into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use frequency density

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find missing frequency

TRIGGER *"work out", "find", a missing value*

03 Read area under a bar

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Build the histogram scale

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use histogram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use frequency table

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 33 website questions, 33 checked solutions, 3 local PDFs read.

01 Use frequency density

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5(40) + 10(20) = 400$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q19

Use the relative histogram heights

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate area

$$5(40) + 10(20) = 400$$

Finish: 370 students.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

02 Find missing frequency

TRIGGER *"work out", "find", a missing value*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5(40) + 10(20) = 400$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

Source pattern: Jan 2020 P1HR Q19

Use the relative histogram heights

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate area

$$5(40) + 10(20) = 400$$

Finish: 370 students.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

03 Read area under a bar

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

1, 4, 9, 6, 1

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q18

Use histogram

1, 4, 9, 6, 1

Use histogram

28, 112, 126, 84, 42

Finish: 4.36 hours.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

04 Build the histogram scale

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM1H Q20

Frequency density

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

Finish: draw bars with densities 2.4, 3.4, 3, 1.4.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

05 Use histogram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 1, 4, 9, 6, 1

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q18

Use histogram

1, 4, 9, 6, 1

Use histogram

28, 112, 126, 84, 42

Finish: 4.36 hours.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

06 Use frequency table

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q21

Use histogram

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

Use frequency table

$$\text{frequency density} = \frac{16}{0.5} = 32$$

Finish: 102 watermelons

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use frequency density
"work out", "find", a missing value	Find missing frequency
"work out", "show", "hence", a familiar exam phrase	Read area under a bar
"work out", "show", "hence", a familiar exam phrase	Build the histogram scale
"work out", "show", "hence", a familiar exam phrase	Use histogram
"work out", "show", "hence", a familiar exam phrase	Use frequency table

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.